Regression Techniques

Introduction to Machine Learning

Module 2

Prof. Rajesh Nag

Compiled by Amitendu and Claude

Contents

1 Introduction to Regression **3**
1.1 Key Terms . . . 3
1.2 Regression vs. Classification . . . 3
1.3 Typical Regression Problems . . . 3
1.4 Families of Regression Models . . . 4
1.5 First Step: Look at the Data . . . 4

2 Simple Linear Regression **5**
2.1 Meaning of the Parameters . . . 5
2.2 Finding the Best Line: Least Squares . . . 5
2.3 Worked Example . . . 6
2.4 How Good Is the Fit? The R^2 Score . . . 6
2.5 Assumptions of Linear Regression . . . 7
2.6 Learning the Line by Gradient Descent . . . 7
2.7 Code . . . 7

3 Multiple Linear Regression (Basic Idea) **9**
3.1 Matrix Form and the Normal Equation . . . 9
3.2 A Geometric Picture . . . 9
3.3 Important Practical Points . . . 9
3.4 Assumptions . . . 10
3.5 Code . . . 10

4 Model Prediction **12**
4.1 From Training to Prediction . . . 12
4.2 Point Predictions . . . 12
4.3 Interpolation vs. Extrapolation . . . 12
4.4 How Uncertain Is a Prediction? . . . 12
4.5 Transformed Targets . . . 13
4.6 Checks Before Trusting Predictions . . . 13
4.7 Code . . . 13

5 Evaluating Regression Models: MAE, MSE and RMSE **14**
5.1 The Error of One Prediction . . . 14
5.2 Mean Absolute Error (MAE) . . . 14
5.3 Mean Squared Error (MSE) . . . 14
5.4 Root Mean Squared Error (RMSE) . . . 14
5.5 Worked Example 1 . . . 15
5.6 Worked Example 2: The Study-Hours Model . . . 15
5.7 Why the Choice Matters: Effect of Outliers . . . 16
5.8 Comparison of the Three Metrics . . . 16
5.9 Other Useful Measures . . . 16
5.10 Evaluation Good Practice . . . 16
5.11 Code . . . 16

6 Revision Toolkit **18**
6.1 One-Page Summary . . . 18
6.2 Formula Sheet . . . 18
6.3 Commonly Confused Pairs . . . 19
6.4 Short-Answer Practice . . . 19

6.5 Long-Answer Outlines 20
6.6 Multiple-Choice Practice 20
6.7 Glossary 21

1 Introduction to Regression

In plain words. Regression answers "*how much?*" questions. If you know a flat's area, can you guess its rent? If you know how many hours a student studied, can you guess the marks? Regression learns the relationship between inputs and a *number* from past examples, then uses it to predict that number for new cases.

Definition (Regression). A supervised learning task in which a model learns a function f from labelled data $\{(x_i, y_i)\}$ so that it can predict a **continuous numerical target** y from one or more input features x.

1.1 Key Terms

Term	Meaning
Independent variable x	Input, predictor or feature (hours studied, area)
Dependent variable y	Output, response or target we want to predict (marks, rent)
Observed value y_i	The actual value in the data
Predicted value $\hat{y}_i$	The value the model outputs (read "y-hat")
Residual (error) e_i	$e_i = y_i - \hat{y}_i$, the gap between actual and predicted
Parameters	Numbers learned from data (slope and intercept)

1.2 Regression vs. Classification

	Regression	Classification
Output	A continuous number	A category
Question	How much? How many?	Which one?
Examples	House price, temperature, marks	Spam or not, disease or healthy
Typical error measure	MAE, MSE, RMSE	Accuracy, precision, recall

1.3 Typical Regression Problems

- Predicting house prices from area, location and age.
- Forecasting sales from advertising spend.
- Estimating a student's marks from study hours and attendance.
- Predicting electricity demand from temperature and time of day.
- Estimating crop yield from rainfall and fertiliser use.

1.4 Families of Regression Models

Model	Idea
Simple linear regression	One input, straight-line relationship (Chapter 2)
Multiple linear regression	Several inputs, linear relationship (Chapter 3)
Polynomial regression	Curved relationship using terms $x, x^2, x^3, \ldots$
Ridge / Lasso regression	Linear regression with a penalty that prevents overfitting
Decision-tree / random-forest regression	Piecewise constant predictions from splits
KNN regression	Average of the target of the k nearest neighbours

This module concentrates on **linear regression**, the foundation of all the others.

1.5 First Step: Look at the Data

Before fitting anything, draw a **scatter plot** of y against x. If the points roughly follow a straight line, linear regression is suitable. The strength of the straight-line relationship is measured by the **correlation coefficient**

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}, \qquad -1 \le r \le 1,$$

where r near $+1$ means a strong upward line, near -1 a strong downward line, and near 0 no linear relationship.

> **Watch out.** Correlation is not causation. Ice-cream sales and sunburn both rise in summer, but ice cream does not cause sunburn. A regression model describes association; it does not prove that changing x will change y.

> **Exam tip.** "Define regression and differentiate it from classification" is a common 5-mark question. Use the two-row table above (output type, example) and one real example of each.

2 Simple Linear Regression

In plain words. Imagine plotting hours studied against marks for several students. The points form a rough upward cloud. Simple linear regression finds the *one straight line* that passes as close as possible to all the points. Once we have that line, we can read off the expected marks for any number of study hours.

Definition (Simple Linear Regression). A regression model with **one** independent variable, assuming a straight-line relationship:

$$y = \beta_0 + \beta_1 x + \varepsilon, \qquad \hat{y} = \beta_0 + \beta_1 x,$$

where β_0 is the **intercept**, β_1 is the **slope**, and ε is random error.

2.1 Meaning of the Parameters

- **Slope β_1:** the average change in y for every one-unit increase in x. If $\beta_1 = 0.6$, each extra hour of study adds about 0.6 marks. A negative slope means y falls as x rises.
- **Intercept β_0:** the predicted value of y when $x = 0$. It may or may not be meaningful in practice (for example, "marks with zero hours of study").

2.2 Finding the Best Line: Least Squares

Many lines could be drawn through the data. We choose the line that makes the **total squared error** as small as possible. This is the **method of least squares**.

$$\text{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i\right)^2.$$

Squaring makes all errors positive, so over- and under-predictions cannot cancel, and it punishes large errors more.

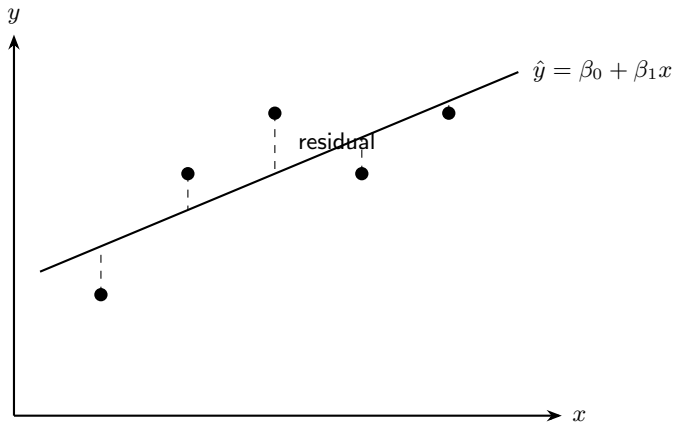


Figure 1: The least-squares line minimises the sum of the squared vertical distances (dashed) from the points to the line.

The formulas

Setting the derivatives of SSE with respect to β_0 and β_1 to zero gives

$$\boxed{\beta_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}, \qquad \beta_0 = \bar{y} - \beta_1 \bar{x}}$$

where $\bar{x}$ and $\bar{y}$ are the means. Two useful facts: the line always passes through the point $(\bar{x}, \bar{y})$, and the residuals of the least-squares line always sum to zero.

2.3 Worked Example

Example: Marks from study hours

Five students: hours $x = 1, 2, 3, 4, 5$; marks (out of 10) $y = 2, 4, 5, 4, 5$.

Step 1: means. $\bar{x} = \frac{15}{5} = 3, \quad \bar{y} = \frac{20}{5} = 4.$

Step 2: deviations.

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
1	2	−2	−2	4	4
2	4	−1	0	0	1
3	5	0	1	0	0
4	4	1	0	0	1
5	5	2	1	2	4
				$S_{xy} = 6$	$S_{xx} = 10$

Step 3: slope and intercept.

$$\beta_1 = \frac{6}{10} = 0.6, \qquad \beta_0 = 4 - 0.6(3) = 2.2.$$

Fitted line: $\hat{y} = 2.2 + 0.6x$. Each extra study hour is associated with about 0.6 extra marks.

Step 4: predictions and residuals.

x	y (actual)	$\hat{y}$ (predicted)	$e = y - \hat{y}$
1	2	2.8	−0.8
2	4	3.4	0.6
3	5	4.0	1.0
4	4	4.6	−0.6
5	5	5.2	−0.2

The residuals sum to $-0.8 + 0.6 + 1.0 - 0.6 - 0.2 = 0$, as expected. SSE $= 0.64 + 0.36 + 1.00 + 0.36 + 0.04 = 2.40$.

2.4 How Good Is the Fit? The R^2 Score

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}, \qquad \text{SST} = \sum (y_i - \bar{y})^2.$$

SST (total variation) is how much y varies around its mean; **SSE** is what the model fails to explain. R^2 is the *fraction of the variation in y explained by the model*: 1 is a perfect fit and 0 means the model is no better than always predicting $\bar{y}$.

Example: R^2 for the study-hours model
SST $= (-2)^2 + 0^2 + 1^2 + 0^2 + 1^2 = 6$ and SSE $= 2.40$, so

$$R^2 = 1 - \frac{2.40}{6} = 0.60.$$

So 60% of the variation in marks is explained by study hours. For simple linear regression $R^2 = r^2$; here $r = \frac{6}{\sqrt{10 \times 6}} = 0.775$ and $r^2 = 0.60$.

2.5 Assumptions of Linear Regression

Assumption	Meaning
Linearity	The relationship between x and y is a straight line
Independence	Observations (and their errors) are independent of each other
Constant variance	The spread of the errors is the same for all values of x (homoscedasticity)
Normal errors	Errors are roughly bell-shaped (needed for confidence intervals and tests, not for fitting)

A **residual plot** (residuals against $\hat{y}$) should look like random scatter around zero; a curve or funnel shape signals a violated assumption.

2.6 Learning the Line by Gradient Descent

Instead of the formula, the same line can be found iteratively. With mean squared error $J = \frac{1}{n}\sum(\hat{y}_i - y_i)^2$:

$$\frac{\partial J}{\partial \beta_1} = \frac{2}{n}\sum(\hat{y}_i - y_i)x_i, \quad \frac{\partial J}{\partial \beta_0} = \frac{2}{n}\sum(\hat{y}_i - y_i), \quad \beta \leftarrow \beta - \alpha\frac{\partial J}{\partial \beta},$$

where α is the learning rate. For one input the closed-form formula is faster, but gradient descent scales to millions of rows and many features.

Example: One gradient-descent step
Data $(1, 2)$ and $(2, 4)$; start $\beta_0 = \beta_1 = 0$; $\alpha = 0.1$. Predictions are $0, 0$ so errors $(\hat{y} - y) = -2, -4$ and $J = \frac{4+16}{2} = 10$.
$\frac{\partial J}{\partial \beta_1} = \frac{2}{2}[(-2)(1) + (-4)(2)] = -10, \quad \frac{\partial J}{\partial \beta_0} = \frac{2}{2}[-2 - 4] = -6.$
Update: $\beta_1 = 0 - 0.1(-10) = 1.0, \quad \beta_0 = 0 - 0.1(-6) = 0.6$. New predictions 1.6 and 2.6 give $J = 1.06$: the error dropped from 10 to 1.06 in one step.

2.7 Code

```
import numpy as np
from sklearn.linear_model import LinearRegression

X = np.array([[1], [2], [3], [4], [5]])      # shape (n_samples, 1)
y = np.array([2, 4, 5, 4, 5])

model = LinearRegression().fit(X, y)
print("slope    :", model.coef_[0])           # 0.6
print("intercept:", model.intercept_)         # 2.2
print("R^2      :", model.score(X, y))        # 0.6
print(model.predict([[6]]))                   # [5.8]
```

Exam tip. Numerical questions give 5 to 8 points and ask for the regression line and a prediction. Always draw the table with $x - \bar{x}$, $y - \bar{y}$, their product and $(x - \bar{x})^2$; write the slope and intercept formulas first; then substitute. Finish with the equation $\hat{y} = \beta_0 + \beta_1 x$.

3 Multiple Linear Regression (Basic Idea)

In plain words. Rent does not depend on area alone. Number of bedrooms, age of the building and locality matter too. Multiple regression simply extends the straight line to *several inputs at once*, giving each input its own "weight" that says how much it moves the prediction.

Definition (Multiple Linear Regression). A regression model with $p \geq 2$ independent variables:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p.$$

Each coefficient β_j is the average change in y for a one-unit increase in x_j, **holding all other variables fixed**.

3.1 Matrix Form and the Normal Equation

For n samples, stack the inputs in a matrix X ($n \times (p+1)$, with a first column of ones for the intercept), the targets in a vector $\mathbf{y}$ and the coefficients in $\boldsymbol{\beta}$:

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Least squares minimises $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ and has the closed-form solution (the **normal equation**)

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

For one variable this reduces to the slope and intercept formulas of Chapter 2. For large data or many features, gradient descent is used instead, since inverting $X^\top X$ becomes expensive.

3.2 A Geometric Picture

With one input the model is a *line*; with two inputs it is a *plane*; with more it is a *hyperplane*, which we cannot draw but which behaves the same way.

Example: House price model

A fitted model (price in lakh rupees; area in sq. ft.; age in years):

$$\hat{y} = 10 + 0.05\,(\text{area}) + 4\,(\text{bedrooms}) - 0.5\,(\text{age}).$$

Reading the coefficients:

- Each extra square foot adds 0.05 lakh, *keeping bedrooms and age the same.*
- Each extra bedroom adds 4 lakh.
- Each extra year of age *reduces* the price by 0.5 lakh.

Predictions:

- House A (1000 sq. ft., 3 bedrooms, 10 years): $10 + 50 + 12 - 5 = 67$ lakh.
- House B (1200 sq. ft., 2 bedrooms, 4 years): $10 + 60 + 8 - 2 = 76$ lakh.

3.3 Important Practical Points

1. Feature scaling

The closed form does not need scaling, but gradient descent converges faster with standardised inputs. Standardising also makes coefficients comparable: a larger absolute standardised

coefficient means a more influential feature.

2. Categorical inputs

Convert categories into 0/1 **dummy variables**. A locality with 3 values (A, B, C) becomes 2 columns ("is_B", "is_C"); category A is the reference. Using all 3 columns together with the intercept causes the *dummy-variable trap* (perfect redundancy).

3. Multicollinearity

When two inputs are highly correlated (area in sq. ft. and area in sq. m.), their individual coefficients become unstable and hard to interpret, even if predictions stay fine. Detect it with the correlation matrix or the **variance inflation factor**

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 comes from regressing x_j on the other inputs. If $R_j^2 = 0.9$ then VIF $= 10$, which is a common warning level. Remedy: drop or combine one of the correlated features.

4. R^2 always rises; use adjusted R^2

Adding any feature, even a useless one, never lowers R^2. The adjusted version penalises extra features:

$$R^2_{\text{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1}.$$

> **Example: Adjusted R^2**
> $n = 30$ samples.
> Model 1: $p = 5$ features, $R^2 = 0.90$. $R^2_{\text{adj}} = 1 - 0.10 \times \frac{29}{24} = 1 - 0.121 = 0.879$.
> Model 2: $p = 10$ features, $R^2 = 0.91$. $R^2_{\text{adj}} = 1 - 0.09 \times \frac{29}{19} = 1 - 0.137 = 0.863$.
> Model 2 has the higher R^2 but the *lower* adjusted R^2: the five extra features were not worth it.

5. Overfitting and regularisation

With many features and little data a model can fit noise. Remedies: feature selection, more data, and **regularisation**, which adds a penalty to the loss:

- **Ridge (L2):** $J = \mathrm{MSE} + \lambda \sum \beta_j^2$ shrinks coefficients.
- **Lasso (L1):** $J = \mathrm{MSE} + \lambda \sum |\beta_j|$ can set some coefficients exactly to zero, performing feature selection.

6. Non-linear relationships

If the data curve, add polynomial terms such as x^2 (polynomial regression) or transform variables (for example, $\log y$). The model is still *linear in the coefficients*, so the same least-squares machinery applies.

3.4 Assumptions

The same four assumptions as simple regression (linearity, independence, constant error variance, normal errors) plus **no severe multicollinearity** and more samples than features.

3.5 Code

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

df = pd.read_csv("houses.csv")
X = df[["area", "bedrooms", "age"]]
y = df["price"]

X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression().fit(X_tr, y_tr)

print(dict(zip(X.columns, model.coef_)), model.intercept_)
print(model.predict(pd.DataFrame([[1000, 3, 10]], columns=X.columns)))
```

> **Exam tip.** Questions on this topic usually ask for the equation, the meaning of coefficients and a prediction. Write the general equation, explain "holding other variables constant", then substitute the given values. Mention multicollinearity and adjusted R^2 to earn full marks.

4 Model Prediction

> **In plain words.** Training finds the line; *prediction* uses it. Once the coefficients are known, predicting is just substituting new input values into the equation. The skill lies in knowing when to trust the answer and when not to.

4.1 From Training to Prediction

1. **Collect and prepare** data (clean, handle missing values, encode categories).
2. **Split** into training and test sets.
3. **Train:** learn $\hat{\beta}$ from the training data only.
4. **Predict** on the test data: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots$
5. **Evaluate** predictions against the true test values (Chapter 5).
6. **Deploy:** use the saved model to predict for brand-new inputs.

The new inputs must be prepared *exactly as the training data was*: the same scaling, the same encodings, the same feature order.

4.2 Point Predictions

> **Example: Predicting with the study-hours line $\hat{y} = 2.2 + 0.6x$**
>
> - $x = 6$ hours: $\hat{y} = 2.2 + 0.6(6) = 5.8$ marks.
> - $x = 2.5$ hours: $\hat{y} = 2.2 + 1.5 = 3.7$ marks.
> - $x = 0$ hours: $\hat{y} = 2.2$ marks (the intercept).

4.3 Interpolation vs. Extrapolation

- **Interpolation:** predicting for x inside the range of the training data ($1 \le x \le 5$). Usually reliable.
- **Extrapolation:** predicting outside that range. Risky, because the straight-line relationship may not continue.

For $x = 20$ hours the line gives $2.2 + 12 = 14.2$ marks, which is impossible when marks are out of 10. The line described the data between 1 and 5 hours only.

> **Watch out.** Never trust a regression prediction far outside the range of the training data. Always check that outputs are physically sensible (no negative prices, no marks above the maximum).

4.4 How Uncertain Is a Prediction?

A regression line gives a single number, but the truth will scatter around it. Typical error is measured by the RMSE (Chapter 5). As a rough guide, for a well-behaved model and plenty of data, about 95% of actual values fall within $\hat{y} \pm 2 \times \text{RMSE}$. For the study-hours model (RMSE ≈ 0.69) the prediction at $x = 6$ is 5.8 ± 1.4, roughly 4.4 to 7.2 marks. (With only five training points this is only an illustration; formal *prediction intervals* account for sample size as well.)

4.5 Transformed Targets

If the model was trained on a transformed target, predictions must be transformed back. If it predicts ln(price) and outputs 4.2, the actual price is $e^{4.2} \approx 66.7$. Likewise, if the target was standardised, apply the inverse of that scaling.

4.6 Checks Before Trusting Predictions

- Are the inputs within the training range?
- Are all features present, in the right units and order?
- Do the residuals on test data look random, with no pattern?
- Does the model beat the simple baseline of always predicting the mean?
- Has the data changed since training (new market, new policy)? If so, retrain.

4.7 Code

```
import numpy as np
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
import joblib

pipe = make_pipeline(StandardScaler(), LinearRegression())
pipe.fit(X_train, y_train)                # scaler and model learn from train only

y_pred = pipe.predict(X_test)             # predictions for the test set
new_house = [[1000, 3, 10]]               # same feature order as training
print(pipe.predict(new_house))

joblib.dump(pipe, "house_model.joblib")   # save for later use
model = joblib.load("house_model.joblib")
```

> **Exam tip.** When asked to "predict", show the substitution step by step and state the unit. If the question gives a value outside the data range, add one line noting that the prediction is an extrapolation and may be unreliable.

5 Evaluating Regression Models: MAE, MSE and RMSE

In plain words. A weather app predicts tomorrow's temperature. How wrong is it, on average? To answer, we look at each prediction's error, then summarise all the errors in a single number. MAE, MSE and RMSE are three ways of doing that summary; they differ in how strongly they react to big mistakes.

5.1 The Error of One Prediction

For sample i, the **error** (residual) is $e_i = y_i - \hat{y}_i$. Positive and negative errors would cancel if simply added, so we remove the sign by taking absolute values or squares.

5.2 Mean Absolute Error (MAE)

Definition (MAE).

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

The average size of the errors, ignoring direction.

- Same units as y and easy to explain: "on average we are off by 0.75 marks".
- Treats every unit of error equally, so it is **robust to outliers**.
- Not smooth at zero, so slightly harder to optimise with gradients.

5.3 Mean Squared Error (MSE)

Definition (MSE).

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

The average of the squared errors.

- Squaring **punishes large errors heavily**: an error of 4 counts sixteen times as much as an error of 1.
- Smooth and differentiable, so it is the standard loss for training linear regression.
- Units are the *square* of the target's units (marks^2), which is hard to interpret directly.

5.4 Root Mean Squared Error (RMSE)

Definition (RMSE).

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

The square root of MSE, which brings the error back to the **original units**.

- Interpretable like MAE but still penalises large errors.

- Always RMSE $\geq$ MAE; the bigger the gap, the more uneven (outlier-heavy) the errors.

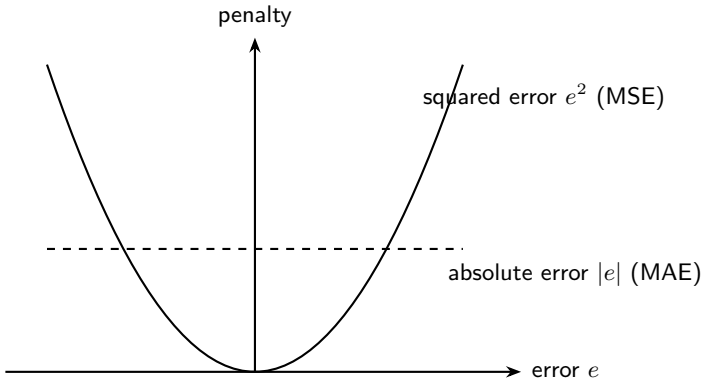


Figure 2: Squared error grows much faster than absolute error as the mistake gets bigger.

5.5 Worked Example 1

Example: Computing MAE, MSE and RMSE

Actual values $y = 3,\ 5,\ 2.5,\ 7$; predictions $\hat{y} = 2.5,\ 5,\ 4,\ 8$.

y	$\hat{y}$	$e = y - \hat{y}$	$\lvert e \rvert$	e^2
3	2.5	0.5	0.5	0.25
5	5	0	0	0
2.5	4	-1.5	1.5	2.25
7	8	-1	1	1
			$\Sigma = 3.0$	$\Sigma = 3.5$

$$\text{MAE} = \frac{3.0}{4} = 0.75, \qquad \text{MSE} = \frac{3.5}{4} = 0.875, \qquad \text{RMSE} = \sqrt{0.875} \approx 0.935.$$

5.6 Worked Example 2: The Study-Hours Model

Example: Training and test errors for $\hat{y} = 2.2 + 0.6x$

Training data (residuals from Chapter 2: $-0.8,\ 0.6,\ 1.0,\ -0.6,\ -0.2$):

$$\text{MAE} = \frac{0.8 + 0.6 + 1.0 + 0.6 + 0.2}{5} = \frac{3.2}{5} = 0.64,$$

$$\text{MSE} = \frac{2.40}{5} = 0.48, \qquad \text{RMSE} = \sqrt{0.48} \approx 0.693.$$

Test data: two unseen students, $x = 6$ (actual 6) and $x = 7$ (actual 6). Predictions: $2.2 + 0.6(6) = 5.8$ and $2.2 + 0.6(7) = 6.4$. Errors: 0.2 and -0.4.

$$\text{MAE} = \frac{0.2 + 0.4}{2} = 0.30, \quad \text{MSE} = \frac{0.04 + 0.16}{2} = 0.10, \quad \text{RMSE} = \sqrt{0.10} \approx 0.316.$$

Baseline check: always predicting the mean $\bar{y} = 4$ gives MSE $= \frac{\text{SST}}{n} = \frac{6}{5} = 1.2$ (RMSE ≈ 1.10). The model's MSE of 0.48 is clearly better, and $R^2 = 1 - \frac{0.48}{1.2} = 0.60$.

5.7 Why the Choice Matters: Effect of Outliers

> **Example: Same MAE, very different RMSE**
>
> Model P makes errors $2, 2, 2, 2$. Model Q makes errors $0, 0, 0, 8$.
> Model P: MAE $= 2$, MSE $= 4$, RMSE $= 2$.
> Model Q: MAE $= \frac{8}{4} = 2$, MSE $= \frac{64}{4} = 16$, RMSE $= 4$.
> Both have MAE $= 2$, yet Q has one large mistake, which RMSE reveals (4 versus 2). If large errors are especially costly (for example, under-predicting hospital demand) prefer RMSE; if all errors matter equally or the data has outliers, prefer MAE.

5.8 Comparison of the Three Metrics

	MAE	MSE	RMSE
Formula	$\frac{1}{n}\sum \lvert e\rvert$	$\frac{1}{n}\sum e^2$	$\sqrt{\frac{1}{n}\sum e^2}$
Units	Same as y	Square of y's units	Same as y
Sensitive to outliers?	Low	High	High
Smooth / differentiable?	No (kink at 0)	Yes	Yes
Common use	Reporting, robust comparison	Training loss, theory	Reporting with emphasis on big errors
Best value	0	0	0

Lower is better for all three. They are only comparable between models evaluated on the *same* target and data; because they depend on the scale of y, an RMSE of 5 means little without knowing whether y is around 10 or around 10,000.

5.9 Other Useful Measures

- R^2**:** $1 - \frac{\text{SSE}}{\text{SST}}$, the fraction of variance explained (scale-free).
- **Adjusted** R^2**:** R^2 penalised for the number of features.
- **MAPE:** $\frac{1}{n}\sum \frac{\lvert y_i - \hat{y}_i\rvert}{\lvert y_i\rvert} \times 100\%$, an error as a percentage. For Example 1: $\frac{1}{4}\left(\frac{0.5}{3} + 0 + \frac{1.5}{2.5} + \frac{1}{7}\right) = \frac{0.1667 + 0.6 + 0.1429}{4} \approx 22.7\%$. It breaks down when some y_i are zero.

5.10 Evaluation Good Practice

1. Compute metrics on the **test set** (or by cross-validation), not only on training data. Training error is optimistic.
2. Compare **training and test errors**: much lower training error signals overfitting; both high signals underfitting.
3. Compare with a **baseline** (predict the mean).
4. Plot **residuals** against predictions; a pattern means the model is missing something.
5. Choose the metric that reflects the real cost of mistakes.

5.11 Code

```
import numpy as np
from sklearn.metrics import (mean_absolute_error, mean_squared_error,
                             r2_score, mean_absolute_percentage_error)

y_true = np.array([3, 5, 2.5, 7])
```

```
y_pred = np.array([2.5, 5, 4, 8])

mae  = mean_absolute_error(y_true, y_pred)          # 0.75
mse  = mean_squared_error(y_true, y_pred)           # 0.875
rmse = np.sqrt(mse)                                 # 0.935
r2   = r2_score(y_true, y_pred)
mape = mean_absolute_percentage_error(y_true, y_pred)   # fraction, 0.227

print(f"MAE={mae:.3f}  MSE={mse:.3f}  RMSE={rmse:.3f}  R2={r2:.3f}")
```

Watch out. Do not forget the square root when asked for RMSE, and do not report MSE as if it were in the original units. Also check the argument order in libraries: it is `metric(y_true, y_pred)`.

Exam tip. The usual numerical: "Given actual and predicted values, calculate MAE, MSE and RMSE." Draw the table of e, $|e|$ and e^2, write each formula, substitute, and give the answers with units. A one-line comment such as "RMSE is larger than MAE because of the larger errors" shows understanding.

6 Revision Toolkit

6.1 One-Page Summary

Topic	Remember this
Regression	Supervised learning with a continuous target; error = actual − predicted.
Simple linear regression	$\hat{y} = \beta_0 + \beta_1 x$; $\beta_1 = \frac{S_{xy}}{S_{xx}}$, $\beta_0 = \bar{y} - \beta_1 \bar{x}$; line passes through $(\bar{x}, \bar{y})$; least squares minimises SSE.
R^2	$1 - \frac{\text{SSE}}{\text{SST}}$: share of variance explained; $R^2 = r^2$ in simple regression.
Multiple regression	$\hat{y} = \beta_0 + \sum \beta_j x_j$; each β_j holds others fixed; $\hat{\beta} = (X^\top X)^{-1} X^\top y$; watch multicollinearity; use adjusted R^2.
Prediction	Substitute inputs; interpolation is safe, extrapolation risky; apply the same preprocessing; invert any target transformation.
MAE / MSE / RMSE	Mean of $\lvert e \rvert$; mean of e^2; $\sqrt{\text{MSE}}$. RMSE $\geq$ MAE; MSE and RMSE punish big errors.

6.2 Formula Sheet

Name	Formula
Slope	$\beta_1 = \dfrac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$
Intercept	$\beta_0 = \bar{y} - \beta_1 \bar{x}$
Correlation	$r = \dfrac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$
SSE, SST	$\sum (y - \hat{y})^2, \quad \sum (y - \bar{y})^2$
R^2	$1 - \dfrac{\text{SSE}}{\text{SST}}$
Adjusted R^2	$1 - (1 - R^2)\dfrac{n - 1}{n - p - 1}$
Normal equation	$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$
Gradient step	$\beta \leftarrow \beta - \alpha \, \partial J / \partial \beta$
MAE	$\frac{1}{n} \sum \lvert y - \hat{y} \rvert$
MSE	$\frac{1}{n} \sum (y - \hat{y})^2$
RMSE	$\sqrt{\text{MSE}}$
VIF	$\dfrac{1}{1 - R_j^2}$

6.3 Commonly Confused Pairs

Pair	First	Second
Regression vs classification	Predicts a number	Predicts a category
Slope vs intercept	Change in y per unit x	Value of y when $x = 0$
Simple vs multiple regression	One input	Two or more inputs
R^2 vs adjusted R^2	Always rises with features	Penalises useless features
Interpolation vs extrapolation	Inside training range	Outside training range
MAE vs RMSE	Treats errors equally	Penalises large errors more
MSE vs RMSE	Squared units	Original units
Correlation vs causation	Association between variables	One variable causes the other

6.4 Short-Answer Practice

1. **Define regression.** Supervised learning of a function that predicts a continuous target from input features.
2. **What is a residual?** The difference between the actual and predicted value, $e = y - \hat{y}$.
3. **State the simple linear regression equation.** $\hat{y} = \beta_0 + \beta_1 x$.
4. **What does the method of least squares minimise?** The sum of squared residuals (SSE).
5. **Give the slope and intercept formulas.** $\beta_1 = S_{xy}/S_{xx}$ and $\beta_0 = \bar{y} - \beta_1 \bar{x}$.
6. **What does a slope of** 0.6 **mean?** y rises by 0.6 on average for each one-unit increase in x.
7. **What does** $R^2 = 0.6$ **mean?** 60% of the variation in y is explained by the model.
8. **State the multiple regression equation.** $\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$.
9. **What is multicollinearity and why is it a problem?** High correlation between inputs, making coefficients unstable and hard to interpret.
10. **Why use adjusted** R^2**?** R^2 never falls when features are added; adjusted R^2 penalises useless features.
11. **Difference between interpolation and extrapolation?** Predicting inside versus outside the range of the training data; the latter is unreliable.
12. **Define MAE, MSE and RMSE.** Mean of absolute errors; mean of squared errors; square root of the mean of squared errors.
13. **Which metric is more sensitive to outliers, MAE or RMSE?** RMSE, because errors are squared.
14. **Why is RMSE preferred to MSE for reporting?** It is in the same units as the target.

6.5 Long-Answer Outlines

Question	Skeleton that earns full marks
Explain simple linear regression with an example	Definition; equation; meaning of β_0, β_1; least squares; formulas; worked numerical with table; fitted line; R^2; assumptions.
Explain multiple linear regression	Equation; interpretation of coefficients; matrix form and normal equation; dummy variables; multicollinearity; adjusted R^2; regularisation; example prediction.
Explain how a regression model makes predictions	Train/test split; substitute inputs; interpolation vs extrapolation; same preprocessing; transformed targets; checks.
Explain MAE, MSE and RMSE with an example	Three formulas; units and outlier behaviour; worked table; comparison; when to use which; outlier example.

6.6 Multiple-Choice Practice

1. Regression predicts: (a) a category (b) a continuous value (c) a cluster (d) a rule
2. The slope in simple linear regression is: (a) S_{xy}/S_{xx} (b) S_{xx}/S_{xy} (c) $\bar{y}/\bar{x}$ (d) r^2
3. The least-squares line always passes through: (a) $(0, 0)$ (b) $(x_{\max}, y_{\max})$ (c) $(\bar{x}, \bar{y})$ (d) the first data point
4. A residual is: (a) $\hat{y} - y$ (b) $y - \hat{y}$ (c) $y + \hat{y}$ (d) $y/\hat{y}$
5. A slope of 0.6 means: (a) y falls 0.6 per unit of x (b) y rises 0.6 per unit of x (c) $y = 0.6$ always (d) $x = 0.6y$
6. Least squares minimises: (a) sum of errors (b) sum of absolute errors (c) sum of squared errors (d) the largest error
7. $R^2 = 1$ means: (a) perfect fit (b) no fit (c) negative correlation (d) overfitting always
8. A multiple regression with 3 inputs has how many coefficients including the intercept? (a) 2 (b) 3 (c) 4 (d) 5
9. Multicollinearity means: (a) too few samples (b) inputs are highly correlated (c) output is categorical (d) errors are normal
10. Adjusted R^2 penalises: (a) extra features (b) large n (c) outliers (d) the intercept
11. Predicting far outside the training range is called: (a) interpolation (b) extrapolation (c) regularisation (d) stratification
12. MAE is the mean of: (a) absolute errors (b) squared errors (c) percentage errors (d) signed errors
13. Which penalises large errors more heavily? (a) MAE (b) MSE (c) both equally (d) neither
14. RMSE has the same units as: (a) y^2 (b) x (c) y (d) R^2
15. Errors $2, 2, 2, 2$ give an RMSE of: (a) 1 (b) 2 (c) 4 (d) 8
16. Errors $0, 0, 0, 8$ give an RMSE of: (a) 2 (b) 4 (c) 8 (d) 16
17. With $\bar{x} = 3$, $\bar{y} = 4$ and slope 0.6 the intercept is: (a) 1.8 (b) 2.2 (c) 4.6 (d) 0.6
18. The line $\hat{y} = 2.2 + 0.6x$ predicts at $x = 4$: (a) 3.4 (b) 4.0 (c) 4.6 (d) 5.2

Note. Answers: 1-b, 2-a, 3-c, 4-b, 5-b, 6-c, 7-a, 8-c, 9-b, 10-a, 11-b, 12-a, 13-b, 14-c, 15-b, 16-b, 17-b, 18-c.

6.7 Glossary

Term	Meaning
Adjusted R^2	R^2 corrected for the number of features
Coefficient	Weight of a feature in the model
Dummy variable	0/1 column representing a category
Extrapolation	Predicting outside the range of training data
Intercept	Predicted y when all inputs are 0
Least squares	Fitting by minimising the sum of squared errors
MAE	Mean absolute error
MSE	Mean squared error
Multicollinearity	Strong correlation among inputs
Normal equation	Closed-form solution $(X^\top X)^{-1} X^\top y$
Regularisation	Penalty that discourages large coefficients
Residual	Actual minus predicted value
RMSE	Square root of the mean squared error
Slope	Change in y per one-unit change in x
SSE / SST	Unexplained / total sum of squares
VIF	Variance inflation factor, a multicollinearity measure

End of Module 2 — Regression Techniques
Introduction to Machine Learning • Prof. Rajesh Nag